M. Awais Aslam

AI/ML Engineer / Data Scientist / Python Developer

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Summary

ML Engineer with 1.5 year of experience in Generative AI, and Python development. Skilled in designing and deploying production-grade AI systems, with strong expertise in Python development, pipeline engineering, and scalable solution delivery.

Experience

Associate Python Developer — Insightlix Pvt Ltd

Jan 2026 – Present

- Brain Tumor Detection System (Django + Deep Learning):
  – Built a deep learning pipeline for brain tumor detection and MRI analysis (edema, lesion segmentation).
  – Applied SynthSeg + NNUNet models for segmentation and generated visual + quantitative outputs for clinical insights.
- AI Real Estate Agent (Conversational AI):
  – Developed an AI chatbot platform with multi-plan selection and user-driven customization.
  – Designed dynamic forms and backend logic for personalized conversational workflows.
- Showtime Validation & Data Classification Automation (FastAPI):
  – Built APIs to extract and validate movie showtimes from multiple sources.
  – Automated Excel validation and amenity classification into standardized formats.
- Set up and configured the PageIndex framework for RAG document indexing and retrieval workflows.
- Deployed and customized OpenClaw AI Assistant solutions for client demo use cases.
- Integrated Stripe for secure payments and subscription workflows in web applications.

Associate Data Scientist — Programmers Force

Sept 2025 – Dec 2025

- Developing computer vision models for Documents Verification (txt extraction & forgery detection) using:
  – OCR, ELA, and pixel-level noise analysis with pre-trained architectures ResNet, EfficientNet, and ViT.
  – Designed ensemble pipelines with parallel model inference for text extraction, and forgery detection.
- Developing models for Facial Verification & Liveness detection:
  – Implemented liveness detection mechanisms to prevent replay and spoofing attacks.
  – Trained and fine-tuned models on large-scale datasets to enhance real-time accuracy and system performance.
- Contributed to the enhancement of Shufti's identity verification system by developing and testing computer vision models for document and facial verification.

Associate AI Developer – Axix Technologies

June 2025 – Aug 2025

- Managed end-to-end backend infrastructure for a Restaurant Order Management System with Django and SQLite, optimizing database performance and security with JWT and RBAC.
- Developed a real-time Communication Tool (Slack-like platform) using Python Flask, implementing WebSocket for live messaging and notifications.
  – Designed RESTful APIs for workspace, channel, and user management with scalable architecture.
  – Integrated chat analyzer(sentiment/trend insights) and next word predictor(LLM-based) models to enhance user engagement.
- Built RESTful APIs with Flask and FastAPI, ensuring seamless integration of AI/ML models into production workflows.

AI Developer Intern — Octaloop Technologies

March 2025 - May 2025

- Developed AI chatbots using FastAPI, MongoDB, and Ollama with LoRA, QLoRA, quantization, and RLHF.
- Built LLM workflows using LangChain tools (LangGraph, LangServe, LangFlow, LangSmith) for prompt design and agent orchestration.
- Created AI agents: Basic Agent, Document-Reader RAG, Financial Analyst, and Video Summarizer RAG Agent.
- Contributed to an AI-Powered Pay-Per-Minute Call Platform Project:
  – Integrated voice agents, payments (Stripe), RBAC, and Twilio call logs via FastAPI.
  – Built agent workflows (custom/knowledge-based) and added OTP authentication.

Machine Learning Intern — Punjab Information Technology Board (PITB)

Dec2024 - Jan2025

- Worked on computer vision and deep learning applications, finds solutions for real-world use cases.
- Gained hands-on experience with OpenCV, YOLO-based object detection models, and OCR techniques for text recognition.
- Optimized object detection and recognition models to enhance accuracy, efficiency, and overall performance.
- Designed and implemented an automatic number plate recognition (ANPR) system.
- Utilized OpenCV for advanced image preprocessing and real-time video stream processing.

Data Science Intern — Ecodecamp

Sep 2024

- Worked on exploratory data analysis , predictive modeling, and customer segmentation initiatives to support data-driven decision-making.
- Strengthened expertise in Python, SQL, and machine learning through hands-on project implementation.
- Applied advanced data science methodologies to solve complex, real-world model problems.
- Developed and delivered actionable insights through comprehensive data visualizations and analytical reports.

Skills

Python | AI | ML | DL | Gen-AI | Computer Vision | Streamlit | Flask | FastAPI | Django | Web Scraping | BeautifulSoup | Selenium | TensorFlow | Scikit-learn | PyTorch | NLP | NLTK | PowerBI | Matplotlib | Seaborn | Plotly | LangChain | LangGraph | LlamaIndex | Agno(Phidata) | RAG | MCP | Fine-Tuning | MongoDB | SQL | ChromaDB | Pinecone | FAISS | Git | GitHub | Docker | Communication | Teamwork

Projects

Communication Tool-like Slack (Flask):

- Developed a Slack-like communication backend using Flask and WebSockets for real-time messaging.
- Implemented RESTful APIs with SQLite and JWT-based authentication for secure user, channel, and workspace management.
- Integrated AI features, including next-word prediction and LLM-based chat analysis.

AI-Powered White Label Pay-Per-Minute Call Platform (FastAPI):

- Contributed to an AI-powered pay-per-minute call platform using FastAPI, integrating voice agents, Stripe payments, RBAC, and Twilio call.
- Built custom and knowledge-based agent workflows and implemented OTP-based authentication for secure access.

Brain Tumor Detection Model (Django):

- Built a deep learning pipeline to detect and analyze brain tumors from MRI scans, including edema and lesion volume extraction.
- Applied SynthSeg for brain segmentation and overlaid results on MRIs to localize tumor-affected regions.
- Produced quantitative insights and visualizations to support clinical interpretation.

Automatic Number Plate Recognition System (Flask):

- Contributed to an ANPR system using Flask and pretrained Hugging Face models for vehicle and number plate detection.
- Implemented vehicle attribute recognition (plate, color, category, model) with SQLite storage.

Restaurant Order Management System (Django):

- Developed a Django-based Restaurant Order Management System with QR code menu access.
- Implemented customer order placement and order routing to the kitchen with order tracking.

MCP-MultiTool AI Agent (LangChain):

- Developed an MCP-based multi-tool AI agent using Google API, supporting dynamic task handling.
- Integrated chatbot, mathematical computation, and weather assistant as callable tools.

Document Reader RAG-Agent (LangChain):

- Developed a LangChain-based RAG agent using the Google Gemini API, enabling users to upload documents and query them conversationally.

Video Summarizer RAG-Agent (Phidata - Agno):

- Built a LangChain-based RAG agent with the Google Gemini API to analyze uploaded videos and answer user queries.

Other Projects:

OpenClaw AI Assistan **|** Real Estate AI Agent **|** PageIndex RAG Chatbot **|** Medical Chatbot **|** Breast Cancer Predictor AI **|** Automation Web-Scrapping **|** Next Word Predictor **|** SMS Spam Classifier **|** Image Classifier **|** Posture Detection **|** Face-Mask Detection **|** Chat Analyzer **|** Match Win Probability System **|** Document Verification System **|** Facail Verfication System **|** Facial liveness Detection

Achievements

- Developed multiple AI agents using frameworks like LangChain, LangGraph, and Phidata-Agno, including real estate chatbot, PageIndex-based system, and OpenClaw assistant, integrated with LLMs for conversational capabilities.
- Fine-tuned a pre-trained transformer-based language model for sentiment analysis using PEFT techniques (LoRA and QLoRA), enhancing performance on a medical dataset for efficient and accurate healthcare applications.
- Designed and implemented robust backend workflows in Python using Flask, FastAPI, and Django to deliver high-performance, secure, and maintainable web applications.
- Handled the backend of a real-time slack like communication tool using Flask and WebSocket, enabling live messaging, workspace/channel management, and secure user authentication.
- Developed scalable backend architectures for AI-powered applications, bridging ML models (e.g., NLP, generative AI) with production workflows.
- Built advanced computer vision models for document and facial verification, contributed to Shufti KYC system, and contribute in the Shufti KYC enhancement team to improve verification and system reliability.
- Developed a brain tumor detection system achieving accurate MRI-based segmentation and analysis, enabling automated clinical insights through deep learning.
- Automated data extraction workflows (web scraping & showtime validation) and implemented Stripe-based payment integration, reducing manual effort and enabling seamless transactions.

Certifications

1. Python for AI **|** Coursera **|** 2023
2. Data Science **|** CampusX **|** 2023
3. Intro to ML/DL **|** CampusX **|** 2024
4. Data Scientist Internship **|** Ecodecamp **|** Sept 2024
5. ML Engineer Internship **|** PITB | Dec 2024
6. AI Developer Internship **|** OctaloopTechnologies **|** March 2025

Education

F.S.C Pre-Engineering
Govt. Degree College
2018-2020

Bachelor's Degree in Computer Science
NCBA&E
Feb 2021 - Jan 2025